Orthemology Notation Gallery (R7B mathematical-typesetting pipeline, Decision 0023)

DRAFT — not peer reviewed.
Review state (R5): fresh-session repository review completed;
not external human peer review; not empirically validated.
Empirical validation not completed: no designed study has been run.
Terminology benchmark not run: every coined term is a candidate; none adopted.
Companion claim status: philosophical conclusions are conditional on stated premises;
creed-internal material is explicitly school-labeled (Atharī) and revelational where stated.

Source revision (git commit): `9deda9c908e40bbf3b3ea5604bf5c3a6683d8afd`
SOURCE_DATE_EPOCH: `1784772365` (UTC 2026-07-23, from the source commit)
Repository: theislampill/orthemology
Build: scripts/build_pdfs.py (markdown-it-py + typst 0.15.0; deterministic; embedded OFL fonts)

Contents

Orthemology notation gallery

Status: DRAFT — not peer reviewed. Rendered by the R7B mathematical- typesetting pipeline (Decision 0023). Every normative symbol in `docs/notation-registry.yaml` appears below as real mathematics: GitHub renders the `$...$` inline math and fenced `math` blocks through MathJax, and `scripts/build_pdfs.py` renders `artifacts/notation-gallery.pdf` through the pinned Typst compiler via `scripts/latex_to_typst_math.py`. This is a **typography layer, not a notation redesign**: each row's meaning is the registry role, unchanged (Decision 0005). `scripts/validate_math_source.py` asserts a bijection between non-retired symbols and this gallery and that `scripts/validate_pdf_math.py` finds no missing-glyph (notdef/tofu) in the rendered PDF — the defect reproduced in `docs/project-closure/r7b/R7B-PDF-MATH-BASELINE.md` (the core `mu`-vector and `C`-vector rendering as notdef under the old `#raw` path). Below they compose correctly as $\vec{\mu}$ and $\vec{C}$.

1. Normative symbols rendered as mathematics

Registry key	Rendered	Role
`A`	A	declared analysis (identifier + version)
`alpha, beta, ...`	α, β	actor indices
`T = task(A)`	$T = \mathrm{task}(A)$	task component of the analysis
`M_univ, O_univ`	$\mathcal{M}, \mathcal{O}$	universal occurrence and ortheme universes
`M_A, O_A`	$\mathcal{M}_A \subseteq \mathcal{M}, \mathcal{O}_A \subseteq \mathcal{O}$	analysis-active domain and repertoire
`Inst_A`	$\mathrm{Inst}_A \subseteq \mathcal{M}_A \times \mathcal{O}_A$	analysis-relative instantiation (the primitive)
`O*(m; A)`	$O^*(m; A)$	true (actual) profile of m under A
`Pi_A`	Π_A	complete-profile space under A
`Pi_A_partial`	Π_A^∂	partial-profile space under A
`C_hat`	$\hat{C}_{A,\alpha,t}(m)$	candidate set of complete profiles
`p_hat`	$\hat{p}_{A,\alpha,t}(m) \in \Pi_A^\partial$	inferred (partial) profile — belief, never ground truth
`Omega`	$\Omega_{A,\alpha,t}$	observation map
`mu`	$\mu = \langle g, S_\mu, \mathrm{select}_\mu, \mathrm{prov}, \mathrm{ver} \rangle$	metaortheme type
`pi_mu`	π_μ	paired meta-policy
`mu_bar`	$\bar{\mu}_{e,j}$	the j-th metaorthemma in episode e
`MetaInst`	$\mathrm{MetaInst}\left(\bar{\mu}_{e,j}, \mu\right)$	token typing of a metaorthemma under a metaortheme
`Gamma_E`	$\Gamma_E = (E, \rightsquigarrow)$	episode DAG
`e_Gamma`	$e_\Gamma = \mathrm{comp}(\Gamma_E)$	composed boundary-level episode
`GoalSchema`	$\mathrm{GoalSchema}(\alpha)$	parametric goal schema
`G_target`	$\mathcal{G}_{\alpha, A_\alpha}$	actor-alpha's grounded target profile set
`Rep_A`	Rep_A	representation family under A
`chi`	$\chi \in \mathrm{Rep}_A$	one representation
`L_A_star`	$L_A^*(\chi)$	best attainable loss/risk under representation chi
`select_mu`	select_μ	metaortheme state-selecting evidence procedure
`estatus_e`	$\mathrm{estatus}_e$	per-claim evidence-status map of episode e
`eps_A`	ε_A	accepted tolerance of the analysis
`theta_stop`	θ_{stop}	ANDON stop/escalation risk threshold
`K_A`	$\mathcal{K}_A$	cause repertoire
`R_A`	$\mathcal{R}_A$	route repertoire
`W_A`	$\mathcal{W}_A$	warrant-state repertoire

`Q_e`	$\mathcal{Q}_e$	claim ledger of episode e
`delta_e`	δ_e	residual-disposition map of episode e
`ReqPath`	$\mathrm{ReqPath}(e)$	governance-derived required pathway verdicts

2. Structural constructs (typesetting showcase)

The pipeline handles the full range of publication constructs the corpus uses. These are the constructs named in the Decision 0023 design spike.

2.1 Episode signature (display)

The formal-core episode signature — the exact site of the reproduced notdef defect — rendered as real mathematics (the vectors $\vec{\mu}$, $\vec{C}$ compose):

$$e = \left\langle \mathrm{id}; m, \kappa, v; x, H; \alpha, w, A, T, t; \vec{\mu}, \mathrm{MetaTok}, \pi; \vec{C}, \hat{p}; r; \mathrm{estatus}; \mathcal{Q}; \delta; \mathrm{hand\_in}, \mathrm{hand\_out}; a, \mathrm{Succ} \right\rangle$$

2.2 Multi-line aligned equations

$$\begin{aligned} \mathrm{TokenAdequate}(\bar{\mu}, e) &\Leftrightarrow \mathrm{MetaInst}(\bar{\mu}, \mu) \wedge \mathrm{Compatible}(\bar{\mu}, A(e)) \\ \mathrm{V3c}(e) &\Leftrightarrow \forall \bar{\mu} \in \mathrm{MetaTok}(e) : \mathrm{TokenAdequate}(\bar{\mu}, e) \end{aligned}$$

2.3 Set comprehension and predicate

Grounded target (a set of **profiles**, $\subseteq \Pi_{A_\alpha}$, not of occurrences): $\mathcal{G}_{\alpha,A_\alpha} = \left\{ p \in \Pi_{A_\alpha} \mid \mathrm{GoalSchema}(\alpha)(p) \right\}$, and strict soundness $\mathrm{StrictlySoundReasoning}_\chi(e) := \mathrm{ReasoningPathAdequate}_\chi(e) \wedge \mathrm{TokenTruthLinked}_\chi(e)$.

2.4 Underbrace

$$\underbrace{O^*(m; A)}_{\text{ground truth, never asserted from discourse}} \neq \underbrace{\hat{p}_{A,\alpha,t}(m)}_{\text{inferred belief}}$$

2.5 Labeled arrows (world transition vs learner update)

$m_t \xrightarrow{a_t} m_{t+1}$ (occurrence lineage) is not $\theta_t \xrightarrow{U_{A(e_t)}} \theta_{t+1}$ (learner update).

2.6 Table containing inline math

Object	Symbol	Non-claim
observation map	$\Omega_{A,\alpha,t}$	not the occurrence
inferred profile	$\hat{p}_{A,\alpha,t}(m)$	not $O^*(m; A)$
metaorthemma	$\bar{\mu}_{e,j}$	not the metaortheme type μ

All notation meanings are those of `docs/notation-registry.yaml` nothing here introduces a new symbol or claim.